

OpenClaw Temperature and Humidity Acquisition Tutorial

Before starting, please complete the peripheral hardware configuration, API Key configuration, model configuration, and three dialogue solution configuration tests in the "Usage Preparation" directory. If you are currently only using TUI / WhatsApp, you can skip the voice interaction configuration. This article assumes the Jetson Nano Docker environment is already prepared and only covers DHT temperature and humidity sensor-related content.

1. Tutorial Objectives

This article demonstrates the natural language control capability of OpenClaw on Jetson Nano through the DHT temperature and humidity sensor. After completing this tutorial, readers will be able to:

- Understand the basic approach for OpenClaw on Jetson to read temperature and humidity
- Read temperature and humidity via WhatsApp, TUI, and voice interaction
- Complete cases including single reading, comfort judgment, threshold alarm, OLED linkage display, and more
- Understand the complete chain from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw Expansion Board
- [Optional] AI Voice Interaction Module + Speaker
- Detail diagram



Note:

- DHT sensor is not suitable for high-frequency continuous reading; it is recommended to wait approximately 2 seconds before reading again.
- If you want to display temperature and humidity on the screen, the OLED also needs to be properly connected.
- If the sensor fails to read data, the module may not be ready yet; try sending the command again.

3. Software Preparation

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Jetson-side entry points used in this tutorial:

```
/root/.openclaw/openclaw.json  
/root/.openclaw/workspace/skills/jetson-hardware-control/SKILL.md  
Corresponding atomic tools (servo/rgb/dht/oled/camera/music)  
/root/yahboom_ws/src/largemodel/main.py
```

Common commands:

```
/usr/local/bin/dht --format json # Read current temperature and humidity, and return  
results  
/usr/local/bin/dht_oled --format json # Read temperature and humidity and display on OLED
```

4. Testing Temperature and Humidity Acquisition

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check. After confirming OpenClaw responds normally, proceed with DHT-specific testing.

Enter TUI by typing the following command in the terminal:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui  
🔧 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.  
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
session agent:main:main  
gateway connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)  
█
```

In the terminal, you can try inputting the following:

- What is the current environmental humidity
- Read current temperature and humidity
- Help me judge if the current environment is comfortable
- Display current temperature and humidity on OLED

```
Read current temperature and humidity

Current temperature and humidity readings:

- Temperature: 29.9°C (85.8°F)
- Humidity: 52%

Sensor: DHT11 ✓

What is the current environmental humidity

Current environmental humidity: 52%

(Temperature is 29.9°C if you need that too) 01F
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
48k/1.0m (5%)
```

5. Three Methods for Dialogue with OpenClaw

Method	Suitable Scenario	Advantages	Recommended Use Case
WhatsApp	Remote viewing of environmental data, classroom demonstration	Convenient on mobile, suitable for multi-person experience	Demonstrate remote natural language reading
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First establish the reading link
Voice Interaction	Voice query, complete AI experience	Closest to real dialogue experience	Final demonstration solution

5.1 WhatsApp Solution

For WhatsApp access, QR code creation, and gateway startup steps, please refer to "Three Dialogue Solution Configuration Tests". This section only includes dialogue examples suitable for the DHT tutorial.

You can directly have conversations, for example:

- Read current temperature and humidity
- Judge if it is currently hot
- If humidity is too low, remind me

19:53



< 2



+86 153 3885 7526 (你)

给自己发消息

OLED

19:52 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.

19:52 ✓✓

Read current temperature and humidity

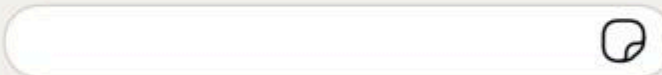
19:53 ✓✓

Current readings from the DHT11 sensor:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 52%

The environment is fairly warm and moderately humid.

19:53 ✓✓



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space

return



If WhatsApp has text replies but no real readings, prioritize check if the reply triggered `/usr/local/bin/dht -format json`.

5.2 TUI Solution

If TUI self-check is already complete, directly type in the container:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔧 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

It is recommended to first test single reading, then test linkage display:

- Read current temperature and humidity
- What is the current temperature
- Give me a brief evaluation based on temperature and humidity
- Read current temperature and humidity and display on OLED

5.3 Voice Interaction Solution

Requires optional AI Voice Interaction Module and speaker.

Execute the following code with openclaw gateway running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

Suitable for directly demonstrating temperature and humidity voice commands, for example:

- What is the current temperature
- Help me read the humidity
- Tell me if it's too hot and humid
- Read current temperature and humidity

```

[系统] ASK=xmfc1_ws/1dc/en_us | 发布=gateway_chat | TTS=ji/xmfc1_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Read current temperature and humidity.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Current readings from the DHT11 sensor:

- **Temperature:** 30°C
  (86°F)
- **Humidity:** 51%

The sensor is working well! 🌡️💧
[信息] 回复摘要失败, 已退回本地清洗播报
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6810 ms 语音, 正在转写

[你] It's too hot. Please, let me know.
[发送] 正在交给 OpenClaw

```

If the voice has recognized the text but no readings, first check whether `openclaw_bridge` has forwarded the voice text to OpenClaw, and whether OpenClaw has generated a DHT action.

6. Comprehensive Cases

The following cases can be executed in TUI, WhatsApp, or voice. It is recommended to use TUI first during the debugging phase.

6.1 Case 1: Single Reading

Experiment Objective

Get current temperature and humidity once to complete the most basic environmental sampling.

Example Commands

- Read current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Case 2: Comfort Judgment

Experiment Objective

Have OpenClaw make a simple judgment of the current environment after reading the data.

Example Commands

- Help me judge if the current environment is comfortable
- Based on current temperature and humidity, is it suitable for prolonged office work?
- Give me a brief evaluation based on temperature and humidity

6.3 Case 3: Threshold Alarm

Experiment Objective

Have OpenClaw give reminders when temperature or humidity exceeds the set range.

Example Commands

- Remind me if temperature exceeds 30 degrees
- If humidity is below 40%,提示空气偏干
- Have temperature and humidity reached alarm conditions now?

6.4 Case 4: OLED Linkage Display

Experiment Objective

Directly display the just-read temperature and humidity results on OLED to form a complete data display chain.

Example Commands

- Read current temperature and humidity and display on OLED
- Write temperature and humidity to the screen
- Make a simple temperature and humidity display page

6.5 Case 5: Weather Station Linkage

Experiment Objective

After reading temperature and humidity, link with OLED and RGB to display environmental status.

Example Commands

- AI weather station
- Read temperature and humidity; if temperature is high, turn on red light; if temperature is suitable, turn on yellow light
- Display temperature and humidity on OLED, and use RGB to indicate temperature status

7. Common Problems

- Temperature and humidity reading failed: Re-initiate the dialogue; if it still doesn't work, execute `dht` in another terminal to confirm if the sensor can read independently.
- OLED not displaying temperature and humidity: First confirm `dht` is normal, then execute `dht_oled`.
- Natural language has replies but no real data: Check if the `dht` tool was called, or if the script uses `/usr/local/bin/dht --format json / /usr/local/bin/dht_oled --format json`.
- Readings occasionally fail: DHT sensor itself is not suitable for high-frequency reading; wait a few seconds and try again; it may also be unstable right after powering on; try sending the command again.
- Voice recognition normal but no action: Check if `openclaw_bridge` and OpenClaw gateway are functioning normally.